

A Semi-Formal Account of the Machine-Checked Odd- N / Dipole Nonemptiness Argument

As Formalized in Isabelle/HOL

Generated from the Isabelle development
Verified_Drone_Theory/Applied_Math_Formalization

June 5, 2026

Abstract

This note reconstructs, in ordinary mathematical language, the antenna / drone-array *nonemptiness* argument exactly as it currently lives in the Isabelle/HOL development. It begins from first definitions (the configuration space, the steered array factor, the radiation pattern, and the robust feasible set whose nonemptiness is the goal), develops the genuinely machine-checked backbone — the Baire-category closeout, the odd- N surjectivity geometry, the explicit 12×12 moment determinant, the steering-transport reduction, and the negligible $\rightarrow$ meager bridge — and ends at the assembled capstone theorems `odd_N_nonemptiness` and `F0_dip_nonempty`. Every named result below is quoted from the Isabelle sources, and *nothing is asserted that is not present there*. A closing section lists, with full transparency, the ten leaf lemmas that are still discharged by `sorry`: the capstone is a machine-checked *assembly* that currently rests on those explicitly named holes. Isabelle identifiers are printed *like this* so each statement can be traced to source.

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1 Status of this document

The development is split into a fully proved *spine* plus a set of *leaf obligations*. Concretely:

- The Baire/meager closeout (§3), the odd- N surjectivity geometry (§4), the explicit moment determinant and steering transport (§5), and the negligible→meager / rank-deficient image engine (§6) are **sorry-free**: their theory files contain no oracle steps.
- The four “bad-stratum” branch results (§7) are proved as *conditional reductions*: each is a genuine theorem whose deep input is named as an explicit hypothesis.
- The two top-level capstones (§8) are assembled and type-checked by Isabelle, but their proofs invoke a handful of leaf lemmas that are presently **sorry**. Those ten leaves are catalogued honestly in §9.

Thus what is *logically complete* is the entire scaffolding plus all the algebra; what is *assembled but not yet airtight* is the very top of the tower. This document is faithful to that state.

2 Background: the objects and the target set

2.1 Configurations and look-directions

Fix an odd number $N \geq 7$ of radiating elements (for the concrete dipole capstone only $N \geq 6$ is needed). In Isabelle $N = \text{CARD}(\text{'n})$ for a finite index type 'n .

Definition 2.1 (Configuration space). A *configuration* is a tuple of N planar element positions

$$x = (x_1, \dots, x_N) \in (\mathbb{R}^2)^N, \quad x \in (\text{real}^2)^{\text{'n}}.$$

This is a finite-dimensional Euclidean space, hence a complete metric (Baire) space — exactly the setting the closeout needs.

A *look-direction* is an angle pair $\omega = (\theta, \phi) \in \mathbb{R}^2$. To each ω one associates a wave-vector $k(\omega)$ with components k_x, k_y, k_z . Steering toward a design direction relative to a steering direction ω_s produces a *planar phase-slope vector* $c(\omega) \in \mathbb{R}^2$. In the concrete (dipole) instance this is the explicit secant map `cvec_dip`:

$$\begin{aligned} c(\omega)_1 &= (k_x(\omega) - k_x(\omega_s)) + \frac{k_x(\omega_0) - k_x(\omega_s)}{k_z(\omega_s) - k_z(\omega_0)} (k_z(\omega) - k_z(\omega_s)), \\ c(\omega)_2 &= (k_y(\omega) - k_y(\omega_s)) + \frac{k_y(\omega_0) - k_y(\omega_s)}{k_z(\omega_s) - k_z(\omega_0)} (k_z(\omega) - k_z(\omega_s)). \end{aligned} \tag{1}$$

The abstract development keeps $c : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ generic; (1) is its instantiation (`cvec_dip`, `cvec0`).

2.2 The array factor and the radiation pattern

Definition 2.2 (Array factor). For a configuration x and direction ω ,

$$A(x, \omega) = \sum_{n=1}^N \exp(-i c(\omega) \cdot x_n) = \sum_{n=1}^N \text{cis}(-c(\omega) \cdot x_n),$$

a sum of N unit-modulus phasors (`af`, `A_cart`). Each summand has modulus 1, so $|A(x, \omega)| \leq N$ always (`cmod_af_le_card`).

Definition 2.3 (Element gain and radiation intensity). Let $g(\omega) \geq 0$ be the (element) gain pattern. The *radiation intensity* (power pattern) is

$$U(x, \omega) = g(\omega) |A(x, \omega)|^2$$

(`Upow`, `U_cart`). For the genuine half-wave *dipole* the gain is $g_{\text{dip}}(\omega) = |e(\theta)|^2$, a smooth function of $\theta = \omega_1$ only (`gain_dip`), and the concrete pattern is $U_{\text{dip}} = \text{U_cart cvec_dip gain_dip}$.

We write $\nabla_{\omega} U$ for the gradient of U in the angle variable (a vector in $\mathbb{R}^2$, `gradU`), and $\nabla_{\omega}^2 U$ for the symmetric 2×2 Hessian (`HessU`); these are the genuine higher derivatives of U supplied by the `HigherDifferentiabilityMulti` library. For a symmetric 2×2 matrix H let

$$\sigma_{\min}(H) = \inf_{\|v\|=1} \|Hv\|$$

be its smallest singular value (`sigma_min`). Finally, the *degeneracy map*

$$\Phi(x, \omega) = (\partial_{\theta} U, \partial_{\phi} U, \det \nabla_{\omega}^2 U) \in \mathbb{R}^3$$

(`Phibad`) vanishes exactly at a *degenerate critical point* of the pattern: $\nabla U = 0$ and $\det \nabla^2 U = 0$ (`Phibad_zero_iff`).

2.3 The feasible set and the robust feasible set

The design constraints carve out a compact feasible set.

Definition 2.4 (Feasible set $\mathcal{F}$). With aperture radius R , minimum spacing $d_{\min}$, focusing geometry (A, B, D) , null direction ω_{null} , main-beam direction ω_0 , null cap δ_{null} and main-beam floor $p_{\min}$,

$$\mathcal{F} = c_{\text{pen}}^{-1}\{0\} \cap \{x : U(x, \omega_{\text{null}}) \in [0, \delta_{\text{null}}]\} \cap \{x : U(x, \omega_0) \in [p_{\min}, g(\omega_0)N^2]\} \cap \overline{B}_R(0),$$

where the spacing penalty $c_{\text{pen}}(x) = \sum_{n \neq m} \max\{0, d_{\min} - \|r'_n - r'_m\|\}$ vanishes iff every focused inter-element distance is $\geq d_{\min}$ (`Ffeas`, `cpen`, `spdist`). Each cut is the preimage of a closed set under a continuous map, so $\mathcal{F}$ is **compact** (`Ffeas_compact`), and around any *strictly* feasible point a whole ball lies in $\mathcal{F}$, so $\mathcal{F}$ has nonempty interior (`ball_inside_Ffeas`).

To obtain a *robust* feasible point — one whose sidelobe structure is quantitatively non-degenerate — one intersects $\mathcal{F}$ with margin sets around the design direction $\text{ctr} = (\theta_0, \phi_0)$.

Definition 2.5 (Robust sets and the target $\mathcal{F}_0$). For radii/margins $\varepsilon, \kappa, \xi > 0$ and the angular scan box $\Omega = [\theta_0 - \frac{\pi}{2}, \theta_0 + \frac{\pi}{2}] \times [\phi_0 - \pi, \phi_0 + \pi]$ (`Omega`, proved compact),

$$X_{\text{rob}}(\varepsilon, \kappa) = \{x : \|\nabla_{\omega} U(x, \omega)\| \geq \kappa \ \forall \omega \in \partial B_{\varepsilon}(\text{ctr})\},$$

$$X_0(\xi, \kappa, \varepsilon) = \left\{x \in X_{\text{rob}} : \|\nabla_{\omega} U(x, y)\| + \sigma_{\min}(\nabla_{\omega}^2 U(x, y)) \geq \xi \ \forall y \in \Omega \setminus B_{\varepsilon}(\text{ctr})\right\},$$

(`Xrobust`, `X0`). The **target set** whose nonemptiness is the flagship result is the *robust feasible set*

$$\boxed{\mathcal{F}_0(\xi, \kappa, \varepsilon) = \mathcal{F} \cap X_0(\xi, \kappa, \varepsilon)} \quad (\text{F0, Fzero}).$$

Remark 2.6. The margin condition $\|\nabla U\| + \sigma_{\min}(\nabla^2 U) \geq \xi$ away from the beam says exactly: at every off-beam direction the pattern is either not critical ($\nabla U \neq 0$) or has a nondegenerate Hessian ($\sigma_{\min} > 0$, i.e. $\det \nabla^2 U \neq 0$). This is the analytic shadow of the degeneracy map Φ of §2, and is precisely what the determinant work secures.

The goal. Prove that for a suitable design and suitable positive margins, $\mathcal{F}_0 \neq \emptyset$.

3 The Baire–category closeout (fully machine-checked)

The strategy is a Baire-category argument: a small number of “bad” parameter sets are each topologically negligible (*meager*), so their union cannot swallow a nonempty open piece of the feasible family, and any leftover point is the desired robust feasible configuration.

Definition 3.1 (Nowhere dense, meager). A set E is *nowhere dense* if $\text{int}(\overline{E}) = \emptyset$ (`nowhere_dense`); a set is *meager* if it is contained in a countable union of nowhere-dense sets (`meager`).

Lemma 3.2 (Stability of meagerness; all `sorry-free`). *Meagerness is preserved by subsets (`meager_subset`), by finite unions (`meager_Un`) and by countable unions (`meager_Union_nat`).*

Lemma 3.3 (Baire). *In a finite-dimensional real normed space, a nonempty open set is not meager:*

$$\text{open } U \wedge U \neq \emptyset \implies \neg \text{meager } U \quad (\text{open_nonempty_not_meager}).$$

These combine into the closeout, which is proved with no oracle steps.

Theorem 3.4 (Closeout, `nonemptiness_from_meager_branches`). *Let V be a nonempty open subset of $\mathcal{F}$, and let B_1, B_2, B_3, B_4 be four “bad” sets such that each $B_i \cap V$ is meager. Suppose every good point lies in some robust margin set, i.e.*

$$x \in V \setminus (B_1 \cup B_2 \cup B_3 \cup B_4) \implies \exists \xi > 0. x \in X_0(\xi).$$

Then $\mathcal{F}_0(\xi) \neq \emptyset$ for some $\xi > 0$.

Proof (as formalized). By Lemma 3.3, V is not meager. The union $\bigcup_i (B_i \cap V)$ is meager (`bad_union_meagerness`, four applications of `meager_Un`). If $V \setminus \bigcup_i B_i$ were empty then V would be a subset of a meager set, hence meager — contradiction. Pick $x \in V \setminus \bigcup_i B_i$; by hypothesis $x \in X_0(\xi)$ for some $\xi > 0$, and $x \in V \subseteq \mathcal{F}$, so $x \in \mathcal{F} \cap X_0(\xi) = \mathcal{F}_0(\xi)$. Hence $\mathcal{F}_0(\xi) \neq \emptyset$ (`final_nonemptiness_from_bad_union`). $\square$

Remark 3.5. A Lebesgue variant is also proved: if each $B_i \cap V$ is merely *negligible* (measure zero) then the same conclusion holds, because a nonempty open set is not negligible (`nonemptiness_from_negligible_branches`).

The four bad sets are the obstruction strata of the pattern; they are defined concretely from the array factor (`Bregzero`, `Bfoldzero`, `Bfoldnonzero`, `Bregnonzero`):

- B_{regzero} : on the regular angular stratum, A fails to be transverse to 0;
- B_{foldzero} : A has a zero on the fold curve Σ ;
- $B_{\text{foldnonzero}}$: a fold-critical point with $A \neq 0$ (reduced to a finite exceptional set E);
- $B_{\text{regnonzero}}$: a degenerate critical point ($\Phi = 0$) with $A \neq 0$ on the regular stratum.

4 The odd- N surjectivity geometry (fully machine-checked)

The very first bad set (A non-transverse to 0) is controlled by a clean parity fact: an array factor zero forces the differential $D_x A : (\mathbb{R}^2)^N \rightarrow \mathbb{C}$ to be onto, *provided N is odd*. This is the formalized core of the paper's Lemma *Azero-surj*.

Lemma 4.1 (Odd parity nucleus). *A sum of an odd number of ± 1 's is odd, hence nonzero (`sum_pm_one_odd_nonzero`). Consequently a real-collinear family of unit phasors $z_n = \varepsilon_n u$ ($\varepsilon_n = \pm 1$, $u \neq 0$) sums to $(\sum \varepsilon_n)u$, whose coefficient has the same parity as the number of terms (`sum_collinear_units_card`).*

Theorem 4.2 (Zero $\Rightarrow$ not collinear, `af_zero_odd_not_collinear`). *If N is odd and $A(x, \omega) = 0$, then the phasors $\text{cis}(-c(\omega) \cdot x_n)$ cannot all lie on one real line through the origin.*

Proof (as formalized). If they were collinear, each would be $\varepsilon_n u$ with $\varepsilon_n = \pm 1$, so $A = (\sum \varepsilon_n)u$. But N odd makes $\sum \varepsilon_n$ odd, hence $\neq 0$, and $u \neq 0$, so $A \neq 0$ — contradicting $A(x, \omega) = 0$. $\square$

Corollary 4.3 (Independent pair, `af_zero_odd_indep_pair`). *Under the same hypotheses two of the phasors are $\mathbb{R}$ -linearly independent, so their real span is all of $\mathbb{C}$ (`complex_real_span_2`).*

Theorem 4.4 ($D_x A$ is onto, `dxA_surj = lem_Azero_surj`). *Let N be odd, $c(\omega) \neq 0$ and $A(x, \omega) = 0$. Then the partial differential*

$$D_x A(x, \omega)[h] = -i \sum_{n=1}^N (c(\omega) \cdot h_n) \text{cis}(-c(\omega) \cdot x_n)$$

is surjective onto $\mathbb{C}$: for every $w \in \mathbb{C}$ there is a displacement h with $D_x A(x, \omega)[h] = w$.

Proof (as formalized). Pick an independent pair of phasors at indices $i \neq j$. Because $c(\omega) \neq 0$, choose d with $c(\omega) \cdot d = 1$; moving only elements i, j along d with real coefficients r, s makes $D_x A[h] = -i(r z_i + s z_j)$. Since z_i, z_j span $\mathbb{C}$ over $\mathbb{R}$, the coefficients (r, s) can be chosen to hit any target w . $\square$

A second elementary spine fact pins down where the steering vector itself vanishes.

Lemma 4.5 (Secant on the sphere, `lem_czero`). *If the design and steering directions are angularly separated ($k_z(\omega_0) \neq k_z(\omega_s)$) and the secant phase-slope vanishes, $c_0(\omega_0, \omega_s, \omega) = 0$, then ω is one of the two distinguished directions: $k(\omega) = k(\omega_0)$ or $k(\omega) = k(\omega_s)$.*

5 The moment map and the explicit 12×12 determinant (fully machine-checked)

The hardest of the four branches ($A \neq 0$, degenerate critical point) is governed by a *moment map* whose differential is generically onto. This is the formalized Lemma *Msurj*, and it is the single largest verified algebraic fact in the development.

Definition 5.1 (Moment map, `M_paper`). For a six-element configuration $x \in (\mathbb{R}^2)^6$ and steering c ,

$$M(x, c) = (A, M_1, M_2, M_{11}, M_{12}, M_{22})(x, c) \in \mathbb{C}^6,$$

the vector of the zeroth, first, and second phase-weighted moments of the array `(A_moment, M1_moment, ..., M22_moment)`. Its Fréchet derivative in the configuration variable is $D_x M(x, c) : (\mathbb{R}^2)^6 \rightarrow \mathbb{C}^6$ (`DM_paper_x`).

Definition 5.2 (The paper’s regular point). At $\kappa = 1$ take the six equally spaced “roots-of-unity” positions

$$x_0 = \left((0, 2), \left(\frac{\pi}{3}, 2\right), \left(\frac{2\pi}{3}, 0\right), (\pi, 0), \left(\frac{4\pi}{3}, 2\right), \left(\frac{5\pi}{3}, 2\right) \right), \quad c_0 = (1, 0)$$

(`x0_paper`, `c0_paper`).

Theorem 5.3 (The explicit determinant, `bigJ_det / bigJ_det_nonzero`). Identifying $\mathbb{C}^6 \cong \mathbb{R}^{12}$, the 12×12 Jacobian $J = D_x M(x_0, c_0)$ has the closed-form determinant

$$\det J = -\frac{5\pi^8}{3} \neq 0.$$

This is proved by an exact symbolic expansion of the explicit numeric/symbolic matrix `bigJ`. Its consequences are all formalized:

Corollary 5.4 (Surjectivity and its open locus). (a) The linear map $v \mapsto Jv$ is onto (`bigJ_surj`), hence $D_x M(x_0, c_0)$ is surjective (`surj_DM_paper_base`).

(b) For any x , surjectivity of $D_x M(x, c_0)$ is equivalent to the transported determinant being nonzero, $m^*(x) = \det(\text{matrix}(MJ_x)) \neq 0$ (`surj_iff_m_star`).

(c) The set $\{x : D_x M(x, c_0) \text{ surjective}\} = \{x : m^*(x) \neq 0\}$ is **open** (`open_surj_stratum`), because m^* is continuous.

To pass from the privileged steering $c_0 = (1, 0)$ to an arbitrary nonzero steering c , the development uses a linear change of variables — the “steering transport” bricks, all `sorry-free`.

Lemma 5.5 (Steering transport, `steering_transport_exists`). For every $c \neq 0$ there is an invertible 2×2 matrix T with $T^\top c = c_0$ (`steering_transport_exists`). Writing T ’s action on configurations as $y \mapsto \text{apply}TT y$, the moments transform covariantly:

$$A(\text{apply}TT y, c) = A(y, c_0), \quad M_k(\text{apply}TT y, c) = \sum_\ell T_{k\ell} M_\ell(y, c_0)$$

and analogously for the second moments M_{11}, M_{12}, M_{22} (the bricks `phase_applyT`, `A_moment_applyT`, `M1_moment_applyT`, ..., `M22_moment_applyT`).

Thus a single explicit regular point at c_0 propagates to a regular point at *every* nonzero steering, and the open regular locus of Corollary 5.4(c) transports along with it. (Turning “one regular point” into “open *dense*” for general c is the rank lower-semicontinuity step `DM_paper_x_open_dense_surjective_gen`, which is one of the remaining leaves; see §9.)

6 The negligible $\rightarrow$ meager engine (fully machine-checked)

A rank-deficient smooth image must be meager; this is the measure-theoretic muscle behind every branch. It is built from a baby Sard theorem and a σ -compact exhaustion.

Lemma 6.1 (Negligible/meager bridge). *A closed negligible set is nowhere dense (`nowhere_dense_closed_negligible`). Nowhere dense implies meager (`meager_nowhere_dense`), and a countable union of closed negligible sets is meager (`meager_negligible_closed_cover`).*

Lemma 6.2 (σ -compact exhaustion, `open_sigma_compact_exhaustion`). *Every open $U \subseteq \mathbb{R}^m$ is a countable union of compact sets.*

Theorem 6.3 (Rank-deficient image is meager, `rank_deficient_C1_image_meager`). *If $U \subseteq \mathbb{R}^m$ is open and $F : U \rightarrow \mathbb{R}^n$ is C^1 with $\text{rank } DF(x) < n$ for all $x \in U$, then $F(U)$ is meager. In particular (`smooth_chart_meager`) a smooth map from an open $U \subseteq \mathbb{R}^m$ with $m < n$ has meager image.*

Proof (as formalized). Exhaust U by compacts K_j ; on each, baby Sard (`baby_sard`) gives that the image $F(K_j)$ is negligible and closed, and `meager_negligible_closed_cover` bundles the countable union into a meager set covering $F(U)$. $\square$

A transversality companion turns a critical/level-set picture into negligibility via countable chart covers (`negligible_critical_values_from_charts`, `countable_chart_cover_of_levelset_2d`, both proved); the single geometric input it still needs — a regular-value chart with the rank-defect bookkeeping — is the keystone leaf of §9. An independent Sard transport result `negligible_singular_image_2n` (in `Sard_Negligible`, sorry-free) supplies the negligibility of the image of a map with non-surjective derivative on the concrete configuration type.

7 The four branches as conditional reductions

Each branch is a genuine, sorry-free theorem; its deep geometric input is named as a hypothesis (and discharged — or still open — elsewhere).

Proposition 7.1 (Regular zero stratum, `prop_regzero`). *Let V be nonempty open and Ω_{reg} open. If the joint map A is C^1 and is jointly transverse to 0 on $V \times \Omega_{\text{reg}}$ (its differential onto wherever $A = 0$ — exactly Theorem 4.4), then*

$$\{x \in V : A(x, \cdot) \text{ not transverse to } 0 \text{ on } \Omega_{\text{reg}}\} \text{ is meager.}$$

Proposition 7.2 (Fold zero stratum, `prop_foldzero`). *A finite union of per-chart meager fold-zero sets is meager: if for each chart $\ell < L$ the slice $\{x \in V : \exists t \in I_\ell, F_\ell(x, t) = 0\}$ is meager, then so is their union over ℓ .*

Lemma 7.3 (Fold fields and the fold curve, `lem_foldfields`). *The kernel and tangent fields along the fold curve are computed explicitly, and $\det J_c(D, \theta, \phi) = (\cos \theta - \sin \theta (D \cdot e_r(\phi))) \sin \theta$, identifying the fold curve Σ as the zero set of this determinant.*

Lemma 7.4 (Finiteness of the exceptional set, `lem_Efinite`). *If g_θ is the real restriction of an entire function G (so $g_\theta(t) = \text{Re } G(t)$), $g_\theta \not\equiv 0$ on a compact interval Θ , and the ϕ -fibres are finite, then the exceptional set $E = \{\omega \in \Sigma : g_\theta(\theta) = 0\}$ is finite.*

Proposition 7.5 (Fold nonzero stratum, `prop_foldnonzero`). *If E is finite and each slice $\{x \in V : F_{\text{crit}}(\omega, x) = 0\}$ ($\omega \in E$) is nowhere dense, then the bad set $\subseteq \bigcup_{\omega \in E} \{x \in V : F_{\text{crit}}(\omega, x) = 0\}$ is meager.*

Proposition 7.6 (Regular nonzero stratum, `prop_regnonzero`). *The degenerate-critical bad set decomposes into four meager pieces:*

$$B_{\text{regnonzero}} \cap V \subseteq (Z_{\text{reg}} \cap V) \cup (Z_{H \equiv 0} \cap V) \cup (B_{\text{CaseB}} \cap V) \cup (B_{H0\text{res}} \cap V),$$

and if each of the four pieces is meager then $B_{\text{regnonzero}} \cap V$ is meager. (The four sub-meagerness facts are the appendix obligations `dim Z`, `Case B`, and the $H \equiv 0$ residue; the bulk of the supporting algebra — codimension cuts, $\Lambda^{(11)}$ residues, nowhere-density of the analytic cuts — is proved sorry-free in `Nonemptiness_Regnonzero_Appendix`, e.g. `prop_dimZ`, `prop_Lambda_common`, `cor_Lambda_closed`.)

8 The assembled capstones

The branches, the feasibility construction, and the robustness soundness are now fed through the sorry-free closeout (Theorem 3.4).

Theorem 8.1 (Abstract capstone, `odd_N_nonemptiness`). *Let N be odd, $N \geq 7$, $c(\omega_N) \neq 0$, Ω_{reg} open with $c \neq 0$ on it, E finite, and each fold slice nowhere dense. Then there is $\xi > 0$ with $\mathcal{F}_0(\xi) \neq \emptyset$.*

The proof literally assembles `branch_regzero_meager`, `branch_foldzero_meager`, `branch_foldnonzero_meager`, `branch_regnonzero_meager`, the feasibility witness, and the robustness soundness through `nonemptiness_from_meager`.

The concrete payoff is for the *actual* dipole pattern, with the real radiation intensity $U_{\text{dip}} = g_{\text{dip}}(\omega)|A(x, \omega)|^2$ and the explicit steered wave-vector c_{dip} of (1) — no abstract c or g .

Theorem 8.2 (Concrete dipole capstone, `F0_dip_nonempty`). *For every $N \geq 6$ there exist a design (A, B, D) , directions $\omega_0, \omega_s, \omega_{\text{null}}, \text{ctr}$, tolerances $R, d_{\text{min}}, \delta_{\text{null}}, p_{\text{min}}$, and positive margins $\xi, \kappa, \varepsilon > 0$ such that the robust feasible set for the dipole pattern is nonempty:*

$$\mathcal{F}_0(c_{\text{dip}}, g_{\text{dip}}, R, d_{\text{min}}, A, B, D, \omega_{\text{null}}, \text{ctr}, \Omega, \delta_{\text{null}}, p_{\text{min}}; \xi, \kappa, \varepsilon) \neq \emptyset.$$

The design is delivered by the construction, not assumed.

Proof (the construction, as formalized). Choose $\omega_0 = (\frac{\pi}{2}, 0)$ (main beam), $\omega_s = (0, 0)$ (steering), $\omega_{\text{null}} = (\pi, 0)$ (null). Then $k_z(\omega_s) \neq k_z(\omega_0)$, and $c_{\text{dip}}(\omega_{\text{null}}) \neq 0$ (its first component is -2). By the roots-of-unity witness `feasible_witness_exists` (using $N > 1$) there is a configuration x with $A(x, \omega_{\text{null}}) = 0$ — the N phases are the N -th roots of unity, whose sum vanishes (`sum_cis_neg_roots_unity`) — and all pairwise spacings ≥ 1 . At the main beam $c_{\text{dip}}(\omega_0) = 0$, so all phasors align and the main-beam power is pinned to its cap $U(x, \omega_0) = g_{\text{dip}}(\omega_0)N^2 > 0$ (`Upow_at_main`, with $g_{\text{dip}}(\omega_0) > 0$ since $\sin \omega_0 = 1$). Picking $p_{\text{min}}, d_{\text{min}}, \delta_{\text{null}}$ to make the witness *strictly* feasible places a whole ball inside $\mathcal{F}$ (`ball_inside_Ffeas`), so $\mathcal{F}$ has nonempty interior. A regular feasible point then exists, and the two Weierstrass continuity conjuncts (continuity of $\omega \mapsto \|\nabla U\|$ and of $\omega \mapsto \sigma_{\text{min}}(\nabla^2 U)$, proved sorry-free in `norm_gradU_dip_continuous_on`, `sigma_min_HessU_dip_continuous_on`) upgrade it to a robust feasible point with positive margins, via `F0_nonempty_of_witness`. $\square$

Remark 8.3. The *assembly* of Theorem 8.2 is machine-checked: the construction of the witness, the nulling, the main-beam pinning, the strict feasibility, the compactness/continuity inputs, and the wiring through the closeout are all genuine proofs. What the proof still *borrow*s are the leaf lemmas listed next.

9 What is not yet airtight: the ten leaf sorrys

In the interest of complete honesty — and in keeping with the rule of this project to never present an oracle step as a proof — here are the exactly ten leaf lemmas currently discharged by `sorry` on which the capstones depend (transitively). Each is a self-contained, clearly delimited geometric or analytic statement; none is hidden inside the assembly.

1. `regular_zero_set_projection_charts_core_2d` (`Parametric_Transversality_Euclidean_Base`, line 371): the regular-value chart with rank-defect bookkeeping — the genuine implicit-function/Sard keystone, to be built from `inverse_function_theorem`.
2. `rank_lower_semicont_open_dense_propagation` (`Nonemptiness_Paper`, line 3835): one regular point of a C^1 map into $\mathbb{C}^6$ propagates to an open *dense* surjective locus (rank lower semicontinuity).
- 3–10. In the frontier file `Nonemptiness_Robust`: the parametric transversality meagerness `parametric_transversality_m` (4474); the general-steering regular point and its open-dense locus `DM_paper_x_regular_point_exists` (4757), `DM_paper_x_open_dense_surjective_gen` (4768); the three $\Phi = 0$ stratum-meagerness lemmas `meager_bad_regular_stratum` (4953), `meager_rank_deficient_stratum` (4967), `meager_steering_singular_stratum` (4980), the $A = 0$ degenerate stratum `meager_Azero_degenerate_stratum` (4994); and the separation margin `no_degenerate_to_sphere_annulus` (5357) that turns pointwise nondegeneracy into a uniform positive margin on a sphere/annulus.

By contrast, everything in §§3–7 — the Baire closeout, the odd- N surjectivity, the $-5\pi^8/3$ determinant and its surjectivity/open-locus consequences, the steering transport, the negligible→meager and rank-deficient-image engine, and the four branch reductions — is `sorry`-free. The mathematical tower is built; the remaining work is to knock out these ten named blocks at the top.

Appendix: dictionary of Isabelle names

Object / result	Isabelle name
Configuration space $(\mathbb{R}^2)^N$	<code>(real^2)^n</code>
Array factor A	<code>af</code> , <code>A_cart</code>
Radiation intensity U	<code>Upow</code> , <code>U_cart</code> , <code>U_dip</code>
Steered wave-vector c	<code>cvec_dip</code> , <code>cvec0</code>
Gradient / Hessian / $\sigma_{\min}$	<code>gradU</code> , <code>HessU</code> , <code>sigma_min</code>
Degeneracy map Φ	<code>Phibad</code>
Feasible set $\mathcal{F}$	<code>Ffeas</code> (compact: <code>Ffeas_compact</code>)
Robust target $\mathcal{F}_0$	<code>F0</code> , <code>Fzero</code>
Baire (open $\Rightarrow$ non-meager)	<code>open_nonempty_not_meager</code>
Closeout	<code>nonemptiness_from_meager_branches</code>
$D_x A$ onto (odd N)	<code>dxA_surj</code> , <code>lem_Azero_surj</code>
$\det J = -5\pi^8/3 \neq 0$	<code>bigJ_det</code> , <code>bigJ_det_nonzero</code>
Surjective $\Leftrightarrow m^* \neq 0$; open locus	<code>surj_iff_m_star</code> , <code>open_surj_stratum</code>
Steering transport	<code>steering_transport_exists</code>
Rank-deficient image meager	<code>rank_deficient_C1_image_meager</code>
Four branch reductions	<code>prop_regzero/foldzero/foldnonzero/regnonzero</code>
Abstract / dipole capstone	<code>odd_N_nonemptiness</code> , <code>F0_dip_nonempty</code>